



Dipayan Sarkar

PHD RESEARCHER · BIOINFORMATICS & COMPUTATIONAL BIOLOGY

Siliguri, West Bengal, India

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🎓 Google Scholar

Research Summary _____ Currently pursuing a PhD in bioinformatics and computational biology, focused on deep learning methods for biological sequence-based prediction and generative model development. Experience includes neural network training, transfer learning with pretrained protein language models, and analysis of large-scale biological datasets. Current work involves autoregressive generative modeling of biological sequences, with ongoing exploration of diffusion-based approaches for biological data.

Work Experience

University of North Bengal

Siliguri, India

PHD RESEARCH SCHOLAR

2022 – Present

- Designed and implemented deep learning architectures for sequence-based protein–protein interaction prediction.
- Developed attention-based hybrid models integrating transfer learning from pretrained protein language models, trained and evaluated on large-scale biological sequence datasets.
- Developed and deployed web-based platforms for model inference and protein interaction network visualization.

Education

University of North Bengal

India

PH.D. IN BIOINFORMATICS

Expected March 2027

Advisor: Dr. Chiranjib Sarkar | Focus: Deep learning-based frameworks for protein–protein interaction network prediction.

University of North Bengal

India

M.Sc. IN BOTANY

2021

Specialization: Biochemistry | 77.25% (First Class)

Ananda Chandra College, University of North Bengal

India

B.Sc. (HONS.) IN BOTANY

2019

63.50% (First Class)

Haldibari High School (H.S.), WBCHSE

West Bengal

HIGHER SECONDARY (CLASS XII)

2016

80.00% (First Class) | Subjects: Physics, Chemistry, Mathematics, Biology, English, Bengali

Haldibari High School (H.S.), WBBSE

West Bengal

SECONDARY (CLASS X)

2014

91.58% (First Class)

Publications

- Sarkar, D., & Sarkar, C. (2025). AttnSeq-PPI: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. *Biochimica et Biophysica Acta (BBA)—Proteins and Proteomics*, 141102.
- Sarkar, C.; Sarkar, D.; Parsad, R.; Mishra, D. (2022). Package EGRNi. CRAN. <https://cran.r-project.org/package=EGRNi>

Skills

Programming	Python (PyTorch, HuggingFace), R (CRAN package dev), C (basic)
Deep Learning	Transformers, attention mechanisms, sequence modeling, transfer learning
Generative AI	Autoregressive sequence generation, diffusion-based approaches, probabilistic modeling
Scalable Training	GPU training, large-scale datasets, parallel workflows on HPC
Bioinformatics	Sequence analysis, sequence-only PPI prediction, generative modeling
Data & Viz	NumPy, Pandas, Matplotlib, Seaborn
Deployment	Docker, Flask, FastAPI, Linux server deployment

Software & Tools

AttnSeq-PPI

DEEP LEARNING-BASED PPI PREDICTION PLATFORM

<https://compbiosysnbu.in/attnseqppi/>

ARACoFusion-PPI

ARABIDOPSIS-SPECIFIC PPI PREDICTION AND NETWORK ANALYSIS TOOL

<https://aracofusion.compbiosysnbu.in/>

Awards & Certificates

June 2021 **CSIR-NET (JRF)**, Life Sciences, All India Rank: 216

India

2022 **GATE**, Life Sciences (XL)

India